Project Documentation for Personal Organizer

Introduction

This project, Personal Organizer, is a simple tool to help people manage tasks, schedule events, and take notes. It is designed to be easy to use while keeping everything organized. The system uses basic programming techniques to make sure it works smoothly and quickly.

The main features of the Personal Organizer are:

1. Task Management:
- Add, remove, and search for tasks.
- Sort tasks based on importance.

2. Event Scheduling:
- Add and view events in the correct order.
- Manage upcoming events easily.

3. Note-Taking:
- Save notes and keep track of changes.
- Undo changes if needed.

This document explains the project and how it works, with simple outlines for the contributions and features.

Outline for Contribution Sheet

Team Contributions

- Yousuf Idrees (23-101077):
  - Worked on Task A (Task Management system to organize tasks) + The Document

- Omar Sayed (23-101091):
  - Worked on Task B (Built the Event Scheduler to arrange events by time) + The Document.

- Yassin Sameh (23-101192):
  - Worked on Task C (Created the Note-Taking system to save and manage notes) + The demo video.

Main Features

1. Task Management

What it does:
- Lets users add tasks with a priority and category.
- Helps users remove tasks or search for them by title or category.
- Displays tasks in order of importance.

How it works:
- Uses a simple linked list to keep tasks organized.

2. Event Scheduling

What it does:
- Allows users to add events with a specific time.
- Shows the next event and lets users remove completed ones.
- Displays all events in the right order.

How it works:
- Uses a min-heap to keep events sorted by time.

3. Note-Taking

What it does:
- Lets users save notes with titles.
- Allows updates to notes and keeps old versions.
- Displays all saved notes.

How it works:
- Uses a hash map for quick access and a stack for version control.

How It Works

1. Tools Used:
- Linked List: For managing tasks.
- Heap: For arranging events by time.
- Hash Map and Stack: For saving and updating notes.

2. User Interaction:
- Simple menus guide the user through adding, removing, and viewing tasks, events, and notes.
- The system ensures everything stays organized and easy to access.

Screenshots

1. Task Manager Interface

- Adding tasks with priority and category.
- Searching for tasks by title or category.

Main Menu:



Task Manager Menu:



Add Task:

```
Task Manager Menu:
1. Add Task
2. Remove Task
3. Search Task by Title
4. Search Tasks by Category
5. Display All Tasks
6. Exit
Enter your choice: 1
Enter task title: Assignment
Enter task category: Uni
Enter task priority: 1
Task Assignment added successfuly
```

Remove Task:

```
Task Manager Menu:
1. Add Task
2. Remove Task
3. Search Task by Title
4. Search Tasks by Category
5. Display All Tasks
6. Exit
Enter your choice: 2
Enter the title of the task to remove: Assignment
Task Assignment removed successfuly!
```

Search Task By Title:

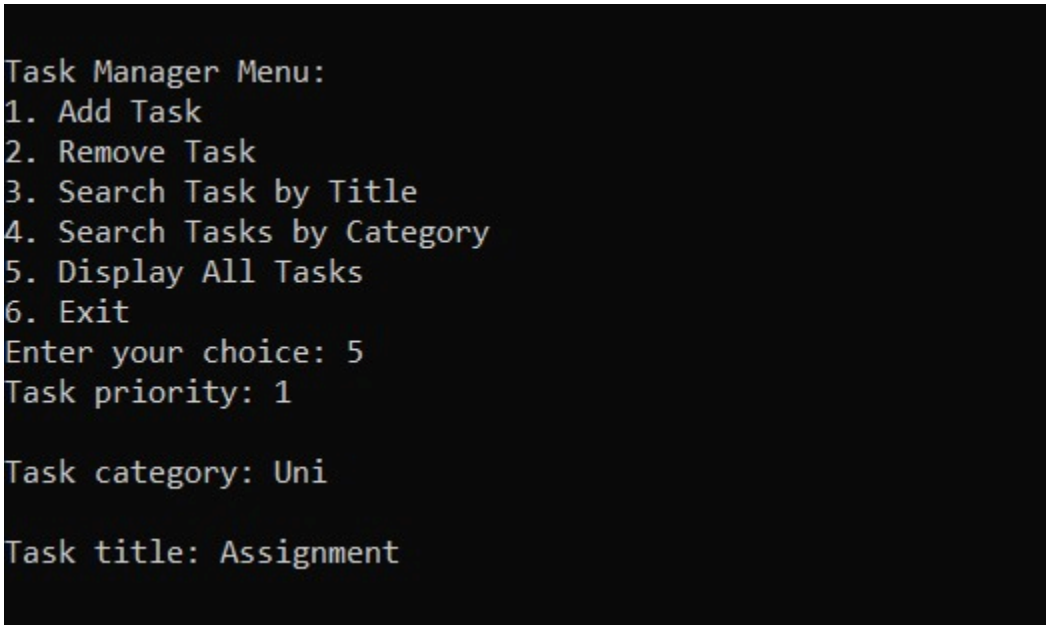
```
Task Manager Menu:
1. Add Task
2. Remove Task
3. Search Task by Title
4. Search Tasks by Category
5. Display All Tasks
6. Exit
Enter your choice: 3
Enter the title of the task to search: Assignment
Task found: Assignment
```

Search Task By Title:

```
Task Manager Menu:
1. Add Task
2. Remove Task
3. Search Task by Title
4. Search Tasks by Category
5. Display All Tasks
6. Exit
Enter your choice: 4
Enter the category to search tasks: Uni
Tasks in category Uni :

Title: Assignment Priority: 1
```

Display Tasks:



2. Event Scheduler Interface

- Adding events with time and description.
- Viewing the next event in order.

Add Event:

```
Event Scheduler Menu:
1. Add Event
2. Remove Next Event
3. View Next Event
4. Display All Events
5. Exit
Enter your choice: 1
Enter event title: Movie
Enter event time (hour 0-23): 12
Enter event time (minute 0-59): 30
Enter event description: El Hareefa
Event added successfully.
```

View Next Event:

```
Event Scheduler Menu:
1. Add Event
2. Remove Next Event
3. View Next Event
4. Display All Events
5. Exit
Enter your choice: 3
Next Event:
Title: Movie
Time: 12:30
Description: El Hareefa
```

Display All Event:

```
Event Scheduler Menu:
1. Add Event
2. Remove Next Event
3. View Next Event
4. Display All Events
5. Exit
Enter your choice: 4
All Events (Closest First):
Title: New year, Time: 12:00, Description: Hang out
Title: Movie, Time: 12:30, Description: El Hareefa
```

Remove Event:

```
Event Scheduler Menu:
1. Add Event
2. Remove Next Event
3. View Next Event
4. Display All Events
5. Exit
Enter your choice: 2
Removing event: Movie
```

3. Note-Taking Interface

- Adding notes and updating them.
- Undoing changes and showing all notes.

Add Note:

```
Note Taking Menu:
1. Add Note
2. Update Note
3. Undo the last update for a note
4. Display all notes
5. Exit
Enter your choice: 1
Enter note title: Training
Enter note content: Go to the training
```

Update Note:

```
Note Taking Menu:
1. Add Note
2. Update Note
3. Undo the last update for a note
4. Display all notes
5. Exit
Enter your choice: 2
Enter the title of the note to update: Training
Enter the new content: Train tomorrow
Note with title 'Training' updated successfully.
```

Undo The Last Update:

```
Note Taking Menu:
1. Add Note
2. Update Note
3. Undo the last update for a note
4. Display all notes
5. Exit
Enter your choice: 3
Enter the title of the note to undo the last update: Training
Undo successful for note with title 'Training'.
```

Display All Notes:

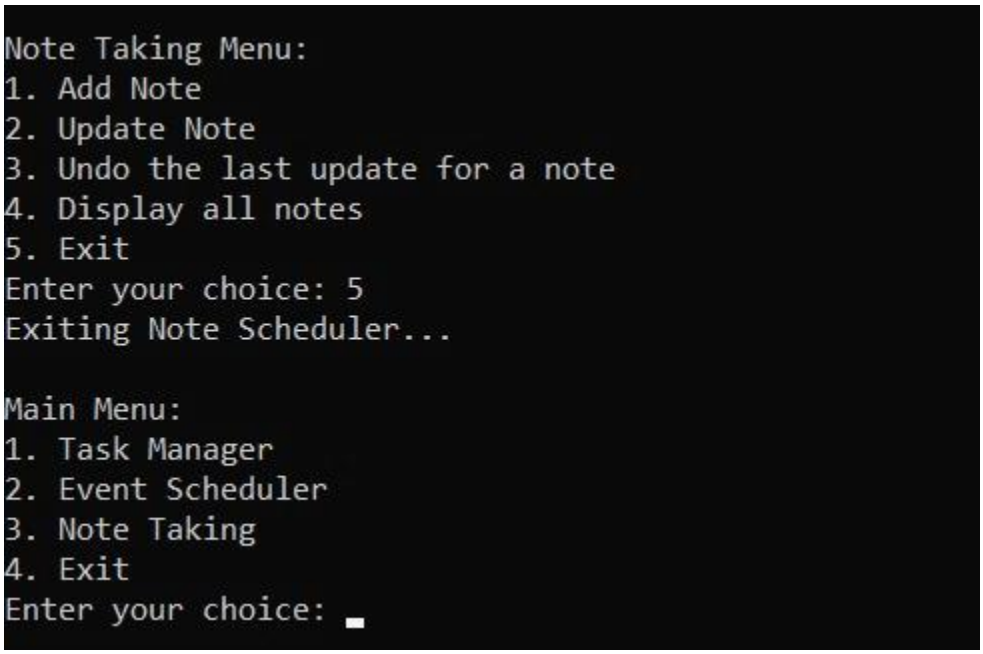
```
Note Taking Menu:
1. Add Note
2. Update Note
3. Undo the last update for a note
4. Display all notes
5. Exit
Enter your choice: 4
 [Title: Training, Content: Go to the training]
```

Conclusion:

The Personal Organizer is a simple and helpful tool for managing daily tasks, events, and notes. It uses basic programming techniques to keep everything in order and easy to use. Future updates can include:

- Saving data for long-term use.
- Adding reminders for tasks and events.
- Allowing more people to use it together.

Exit To Main Menu:



Exit The Program:

